

Mathematics

Stage 8

Paper 1

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You are **not** allowed to use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Write down the coordinates of two possible points that are on the line $y = x + 4$

(..... ,) and (..... ,) [1]

- 2 Write **integers** in the boxes to make each statement correct.

$$7^0 = \boxed{}$$

$$7^{20} \div 7^{18} = \boxed{}$$

[2]

- 3 Work out.

$$\sqrt[3]{-1000}$$

..... [1]

- 4 Draw a line to match each formula, equation or expression to the correct description.

$$3x + 4 = 19$$

Formula

$$10mp$$

Equation

$$A = b \times h$$

Expression

$$\frac{y}{4} = 20$$

[1]

- 5 Complete this sentence.

35 miles is equal to km. [1]

- 6 (a) Write down the gradient of the line $y = 3x + 5$

..... [1]

- (b) Write down the y -intercept of the line $y = 7 - 8x$

..... [1]

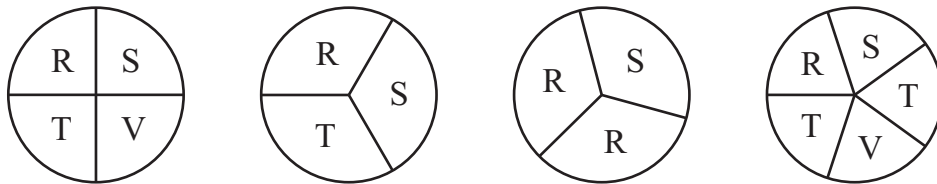
7 Work out.

$$8\frac{4}{7} - 3\frac{2}{7}$$

Give your answer as a mixed number in its simplest form.

..... [2]

8 When a fair spinner is spun the arrow is equally likely to point in any direction.
Chen has these four fair spinners.



He spins one of the spinners 600 times and the arrow points to the letter R 205 times.

Draw a ring around the spinner he is most likely to have used.

[1]

9 Complete each sentence with the correct number.

A regular octagon has lines of symmetry.

A regular hexagon has rotational symmetry of order

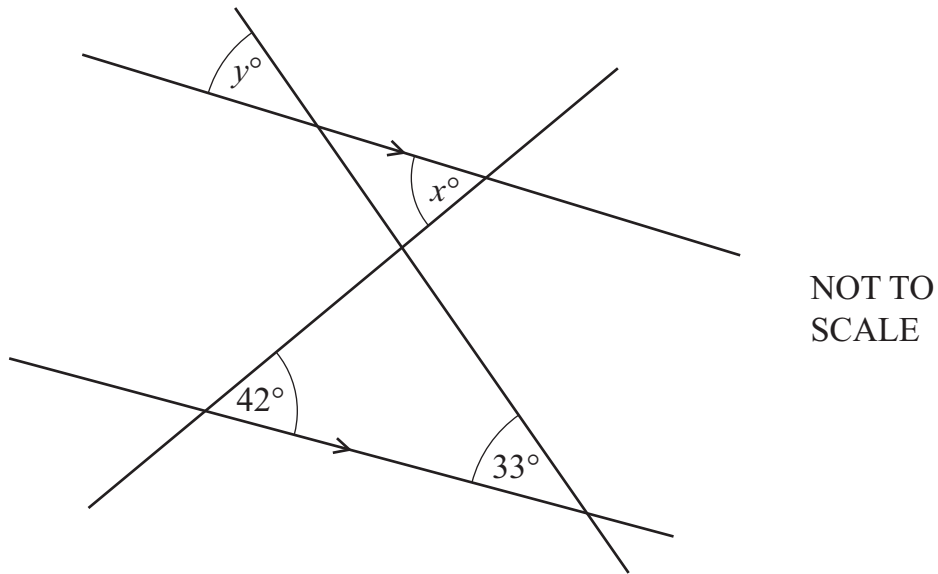
[1]

10 Factorise.

$$6x + 24$$

..... [1]

11 The diagram shows two straight lines crossing two parallel lines.



Find the value of x and the value of y .
Give a geometrical reason for each answer.

$x =$ because

$y =$ because

[2]

- 12** Complete the grid so that the three numbers in each row, each column and each diagonal have the same total.

		-12
-8	-1	6
	-16	

[2]

- 13** Find the value of each of these expressions when $x = -8$

$$10x^2$$

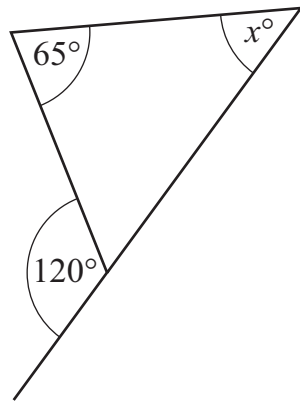
.....

$$5(12 - x)$$

.....

[2]

- 14 The diagram shows a triangle with an exterior angle of 120° .



NOT TO
SCALE

Work out the value of x .

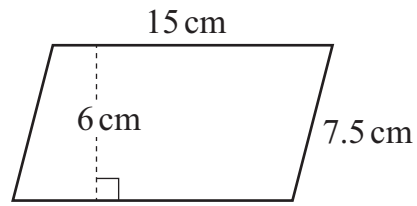
$x =$ [1]

- 15 Expand the brackets.

$$5t(t^2 + 2t - 3)$$

..... [1]

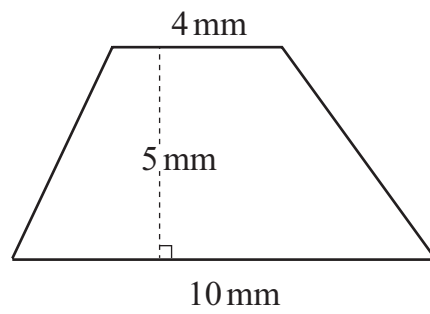
16 (a) Work out the area of this parallelogram.



NOT TO
SCALE

..... cm^2 [1]

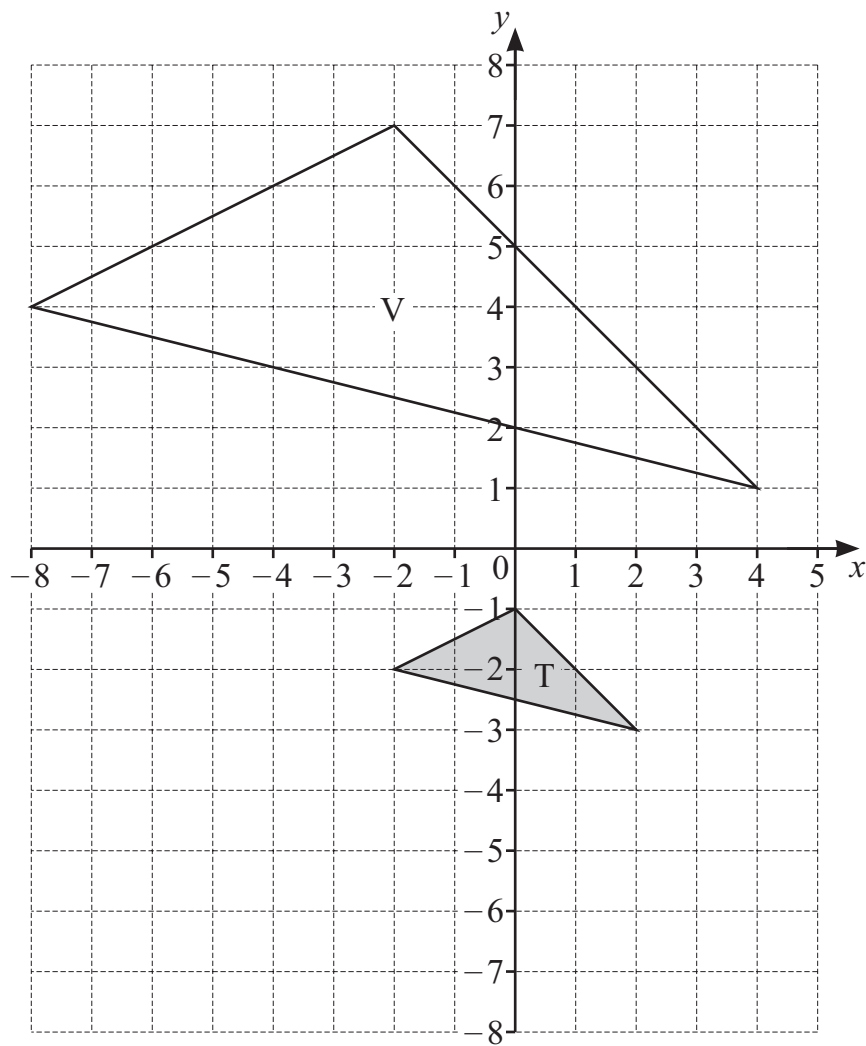
(b) Work out the area of this trapezium.



NOT TO
SCALE

..... mm^2 [2]

17 The diagram shows two triangles, T and V.



(a) A transformation maps triangle T onto triangle V.

Complete these sentences.

The type of transformation is

The scale factor is

[2]

(b) Rotate triangle T by 270° anticlockwise about the point $(-2, -3)$.

[2]

18 Mike has these four number cards.



He uses each card once to make a four-digit number.

Work out how many different four-digit numbers he could make.

..... [1]

19 Jamila has p pencils.

Hassan has 3 fewer pencils than Jamila.

Naomi has 4 times as many pencils as Hassan.

Write an expression, in terms of p , for the total number of pencils the three children have.

Write your expression in its simplest form.

..... [3]

20 Here is a table showing information about some linear number sequences.

Sequence name	First four terms of sequence	How the sequence is related to sequence A	Term-to-term rule	5th term
A	6, 10, 14, 18, ...		Add 4	22
B	7, 11, 15, 19, ...	Add 1 to each term in sequence A	Add 4	23
C			Add 4	20
D		Double each term in sequence A		
E	-6, -10, -14, -18, ...			

Complete the table.

[3]

21 n lies in the interval $3.5 < n < 3\frac{9}{16}$

Find a possible value of n .

Give your answer as a mixed number.

$n =$ [1]

22 Draw a ring around the inequality that is equivalent to $x \leq 4$

$x - 1 \leq 5$

$2x < 8$

$x < 5$

$x + 1 \leq 5$

$x - 3 \geq 1$

[1]

23 Work out.

$$11^2 - 5 \times 2.6 + \sqrt{9^2 + 63}$$

..... [2]

24 Rajiv has two pieces of string measuring 240 cm and 168 cm in length.
He cuts both pieces of string into smaller pieces that are all equal in length.

Work out the greatest possible length for each smaller piece.

..... cm [3]

- 25** In June, eleven people take part in a long jump competition.
Here are the results for the distance, in metres, each person jumps.

2.6 2.4 3.6 2.8 4.2 3.8 5.1 3.0 4.5 3.3 4.6

- (a)** Complete the stem-and-leaf diagram for this information.

2					
3					
4					
5					

Key:

[3]

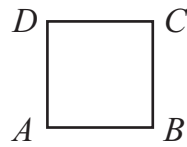
- (b)** In August, the same eleven people take part in another long jump competition.
The median distance jumped in August is 4.1 m.

Write down a comparison between the distances jumped in June and the distances jumped in August.

You must use statistics to support your comparison.

.....
..... [2]

26 Here is square $ABCD$.



$ABCD$ is drawn on a grid so that A is at $(2, 1)$ and C is at $(6, 5)$.
 $ABCD$ is **reflected** in the y -axis to make square $A'B'C'D'$.

Write each of these points in the correct place in the table.

$(3, -3)$ $(-2, 3)$ $(-5, 6)$ $(-3, 3)$ $(-2, 5)$

Inside square $A'B'C'D'$	Outside square $A'B'C'D'$	A vertex of square $A'B'C'D'$	On the edge of square $A'B'C'D'$ but not a vertex

[2]

27 Lily thinks of a number.

She squares the number and the answer is 4.6225

There are two possible numbers she could be thinking of.

Write down the **sum** of these two numbers.

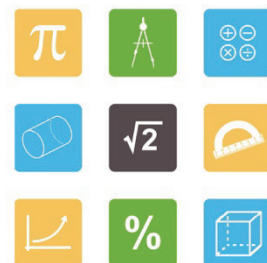
..... [1]

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced annually and is available to download at <https://lowersecondary.cambridgeinternational.org/>

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.



Mathematics

Stage 8

Paper 1

2023

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Brackets in mark scheme

When brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6(58...)	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30.....cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e. 18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

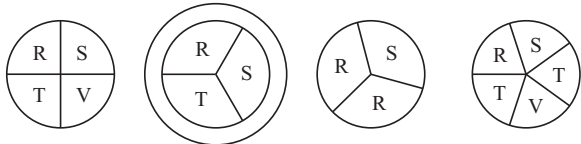
For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance
1	Any two pairs of coordinates with coordinates $(k, k + 4)$.	1		k does not need to be an integer.
2	1 and 49	2	Award 1 mark for each correct answer.	Do not accept 7^2
3	-10	1		
4	$3x + 4 = 19$ $10mp$ $A = b \times h$ $\frac{y}{4} = 20$ Formula Equation Expression	1		Accept any clear indication. All four lines correct for the mark.
5	56	1		Accept 56.3...
6(a)	3	1		
6(b)	7	1		

Question	Answer	Marks	Part Marks	Guidance
7	$5\frac{2}{7}$ correct answer only	2	Award 1 mark for $\frac{37}{7}$ or correct mixed number not in simplest form or correct starting point, e.g. $5\frac{4}{7} - \frac{2}{7}$, $8\frac{2}{7} - 3$	Accept less efficient methods, e.g. $\frac{60}{7} - \frac{23}{7}$
8		1		Accept any clear indication.
9	8 and 6	1		Both sentences correct for the mark.
10	$6(x + 4)$ final answer	1		Do not accept partially factorised, e.g. $3(2x + 8)$
11	$(x =) 42$ (because) alternate angles and $(y =) 33$ (because) corresponding angles	2	Award 1 mark for one correct sentence or 42 and 33 correct.	Accept omission of 'angles'.

Question	Answer	Marks	Part Marks	Guidance									
12	<table><tr><td>-5</td><td>14</td><td>(-12)</td></tr><tr><td>(-8)</td><td>(-1)</td><td>(6)</td></tr><tr><td>10</td><td>(-16)</td><td>3</td></tr></table>	-5	14	(-12)	(-8)	(-1)	(6)	10	(-16)	3	2	Award 1 mark for one row, column or diagonal total equal to -3 seen or implied.	1 mark implied by 14, 3 or 10 correct.
-5	14	(-12)											
(-8)	(-1)	(6)											
10	(-16)	3											
13	640 and 100	2	Award 1 mark for each correct answer.										
14	(x =)55	1											
15	5t ³ + 10t ² – 15t final answer	1											
16(a)	90 (cm ²)	1											
16(b)	35 (mm ²)	2	Award 1 mark for a correct method, e.g. $\frac{1}{2} \times 5 \times (4 + 10)$	Accept in stages, e.g. if split into rectangle and triangles where the sum of the base lengths of the triangles is 6									

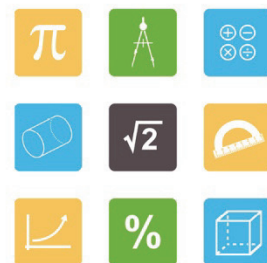
Question	Answer	Marks	Part Marks	Guidance
17(a)	enlargement and 3	2	Award 1 mark for each correct sentence.	
17(b)	Triangle at $(-1, -3)$, $(0, -5)$, $(-2, -7)$.	2	Award 1 mark for correct size and orientation but wrong position or correct rotation of 270° clockwise.	270° clockwise triangle is at $(-3, -3)$, $(-2, 1)$, $(-4, -1)$.
18	4	1		
19	$6p - 15$ or $3(2p - 5)$ final answer	3	Award 2 marks for correct unsimplified answer seen, e.g. $p + p - 3 + 4(p - 3)$ or Award 1 mark for $p - 3$ or $4 \times \text{their}(p - 3)$.	

Question	Answer					Marks	Part Marks	Guidance
20	Name	Sequence	Related to A	Rule	5th term	3	Award 1 mark for each correct row.	Accept equivalent statements in column 3, e.g. take 2 from each term (in sequence A). Accept for E column 3 'make each term negative'. Ignore extra values in column 2
	A	6, 10, 14, 18, ...		Add 4	22			
	B	7, 11, 15, 19, ...	Add 1 to each term in sequence A	Add 4	23			
	C	4, 8, 12, 16, ...	Subtract 2 from each term (in sequence A)	Add 4	20			
	D	12, 20, 28, 36, ...	Double each term in sequence A	Add 8	44			
	E	-6, -10, -14, -18, ...	Multiply each term (in sequence A) by -1	Subtract 4	-22			
21	Any mixed number n where $3.5 < n < 3\frac{9}{16}$ e.g. $3\frac{17}{32}$					1		
22	$x + 1 \leq 5$					1		Accept any clear indication.
23	120					2	Award 1 mark for two correct out of 121 (-) $\sqrt{144}$ $\sqrt{144}$ or 12	

Question	Answer	Marks	Part Marks	Guidance
24	24 (cm)	3	Award 2 marks for correct method to find HCF, e.g. 240 and 168 correctly written as a product of primes or equivalent or answer $2^3 \times 3$ $240 = 2^4 \times 3 \times 5$ $168 = 2^3 \times 3 \times 7$ or Award 1 mark for 240 or 168 correctly written as a product of primes.	Accept factor trees or repeated division leading to correct prime factors instead of product of primes.

Question	Answer	Marks	Part Marks	Guidance								
25(a)	<table><tr><td>2</td><td>4 6 8</td></tr><tr><td>3</td><td>0 3 6 8</td></tr><tr><td>4</td><td>2 5 6</td></tr><tr><td>5</td><td>1</td></tr></table> Key 2 4 represents 2.4 m	2	4 6 8	3	0 3 6 8	4	2 5 6	5	1	3	Award 2 marks for a stem-and-leaf diagram with all 11 numbers in correct numerical order. or Award 1 mark for correct leaves but not in order or for two correct rows and Award 1 mark for correct key.	Any appropriate numbers may be used in the key (does not have to be 2 and 4). Accept equivalents, e.g. “=” in place of represents. The decimal point must be in the key, not in the stems.
2	4 6 8											
3	0 3 6 8											
4	2 5 6											
5	1											
25(b)	Correct comparison with supporting value, e.g. The people jumped further (on average) in August as the median in June is 3.6 m.	2	Award 1 mark for an incomplete comparison, e.g. • The median was lower in June. • They jumped further (on average) in August. • In June the median was 3.6 m (but in August it was 4.1 m). Follow through from the median from <i>their</i> diagram for 1 or 2 marks.	1 mark answers include: • Correct comparison with no median stated. • Correct median for June stated with no comparison (‘but’ is not a comparative word).								

Question	Answer				Marks	Part Marks	Guidance
26	Inside	Outside	A vertex	On the edge	2	Award 1 mark for three or four coordinates correctly placed.	Coordinates in Column 2 can be in either order.
	(-3, 3)	(3, -3) (-5, 6)	(-2, 5)	(-2, 3)			
27	0				1		



Mathematics

Stage 8

Paper 2

2023

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Calculator
Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You may use a calculator.

INFORMATION

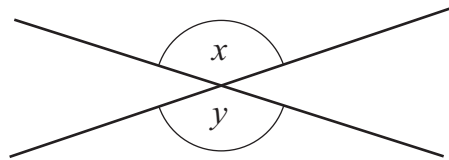
- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 The value of Eva's car is \$9000
This value decreases by 15%.

Work out the new value of Eva's car.

\$ [2]

- 2 The diagram shows two vertically opposite angles formed where two lines intersect.



Angle x and angle y are obtuse angles.

Write down one possible pair of values for angle x and angle y .

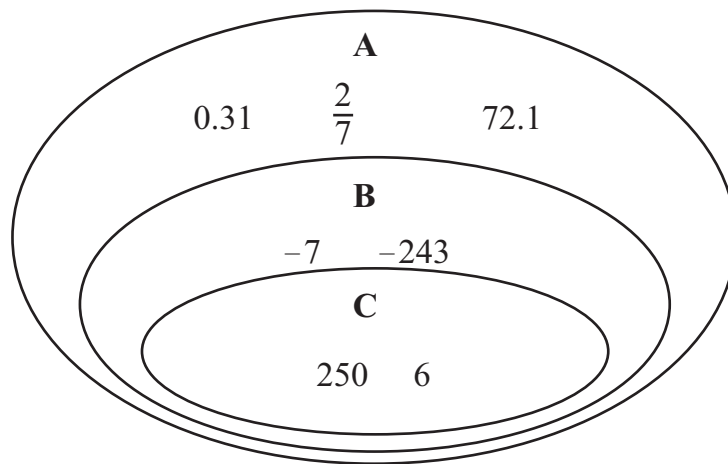
$x =$ °

$y =$ °

[1]

3

3 Here is a Venn diagram with sets A, B and C.



Choose the correct words from this list to describe each of the sets.

natural numbers

rational numbers

integers

A =

B =

C =

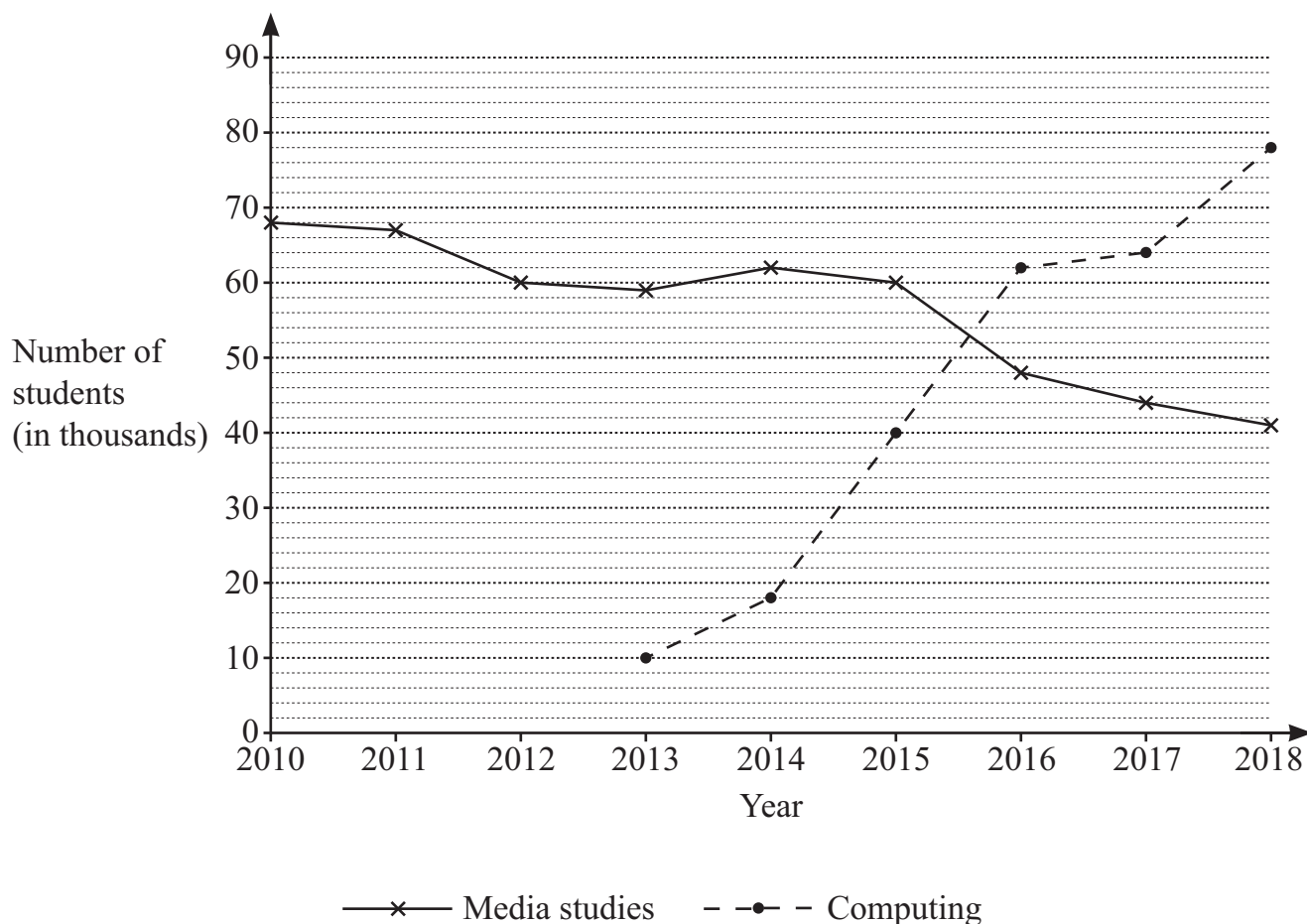
[1]

4 Draw a bearing of 075° from point X.



[1]

- 5 The time-series graph shows information about the number of students studying media studies or computing.



Complete these sentences.

The first one has been done for you.

The trend in the number of students studying media studies is decreasing.

The computing course was first studied in the year 2013.

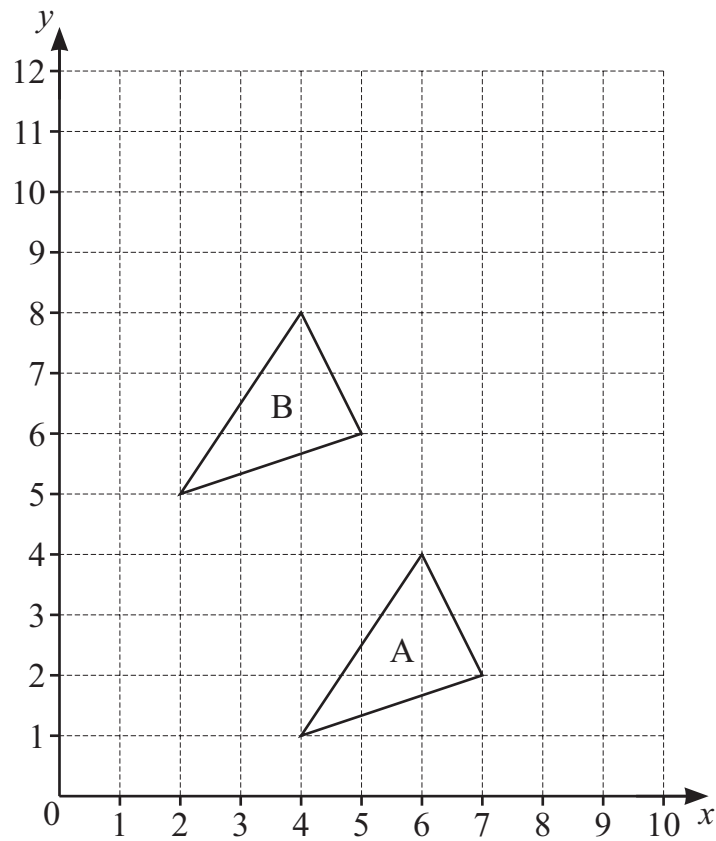
The trend in the number of students studying computing is increasing.

In the year 2015 there were 20 thousand more students studying media studies than computing.

In the year 2018 there were 37 000 more students studying computing than media studies.

[2]

- 6 The diagram shows two triangles drawn on a grid.



- (a) Write down the vector that translates triangle A onto triangle B.

$$\begin{pmatrix} \\ \end{pmatrix}$$

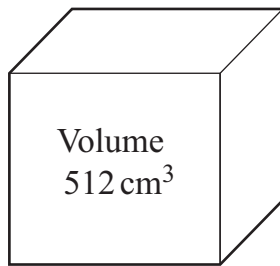
[1]

- (b) On the grid, draw the image of **triangle B** after a translation by vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$.

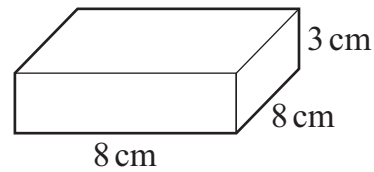
[1]

- 7 Here is some information about three shapes made using cubes and cuboids.

Shape A: cube

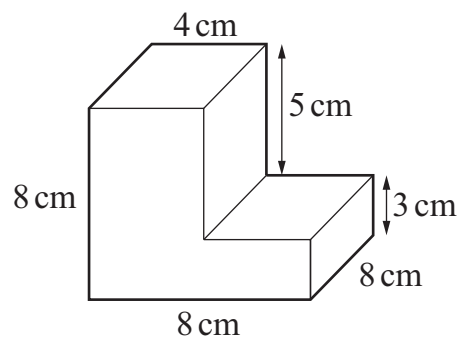


Shape B: cuboid



Shape C:
made by joining shape B
to another cuboid

NOT TO
SCALE



Tick (✓) to show if each of the statements is true or false.

	True	False
Shape A and shape C have the same height, 8 cm.	☐	☐
Shape C has a greater surface area than shape B.	☐	☐
Shape A has the greatest volume.	☐	☐
Shape B has the smallest volume.	☐	☐

[2]

8 Write each ratio in its simplest form.

(a) $550 \text{ cm} : 2.25 \text{ m}$

..... : [2]

(b) $115 : 46 : 161$

..... : : [1]

9 Rearrange each formula to make x the subject.

(a) $r = x + 5$

$x =$ [1]

(b) $y = \frac{7+x}{2p+3}$

$x =$ [2]

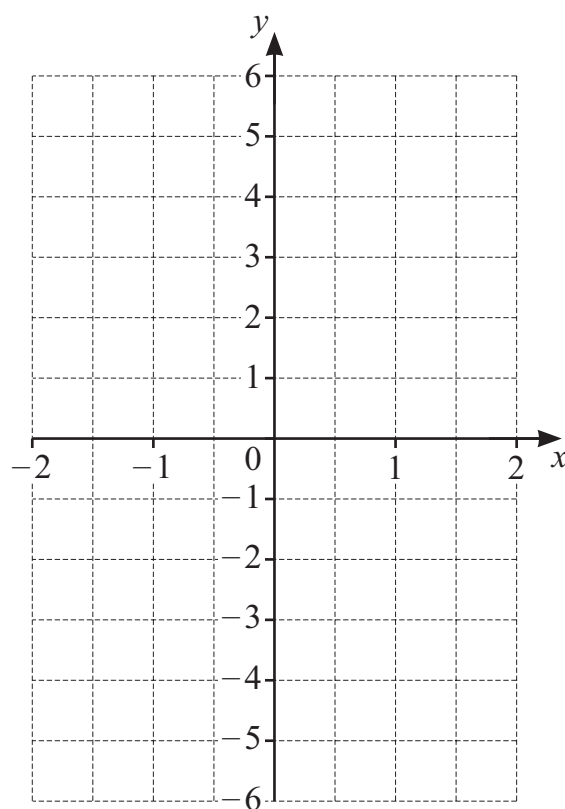
10 Complete the table.

Number	Number given correct to 3 significant figures
806 317	
0.070054	
3.995	

[2]

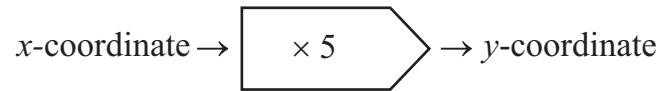
11 Complete the table for $y = 1 - 2x$ and draw the graph of $y = 1 - 2x$ on the grid.

x	-2		2
y		3	



[3]

- 12** Angelique uses this function machine to work out coordinates of points on a line.



Write each of these coordinates in the correct place in the table.

One has been done for you.

(5, 1) (-10, 50) (-1, -5) (5, 10) (2.5, 12.5) (2, 10)

	On Angelique's line	Not on Angelique's line
Below the line $y = 10$		(5, 1)
On the line $y = 10$		
Above the line $y = 10$		

[2]

- 13** The angles in a quadrilateral are in the ratio 3 : 6 : 7 : 2

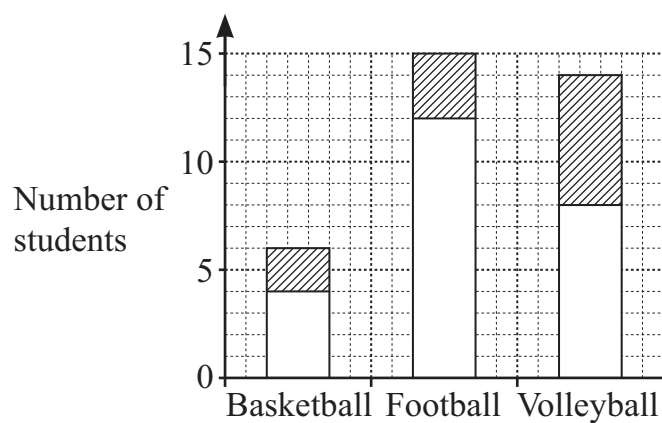
Work out the size of the largest angle in this quadrilateral.

..... ° [2]

14 The table shows the favourite sports for a group of students.

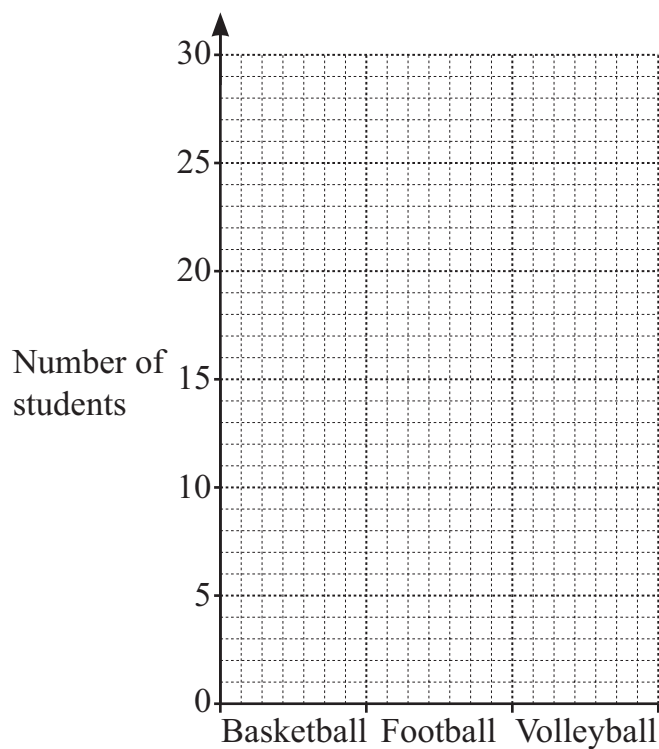
	Basketball	Football	Volleyball
Number of girls	4	12	8
Number of boys	6	15	14

Safia draws this compound bar chart to represent the information.



Safia has made some mistakes.

Draw the compound bar chart correctly.



- 15** Oliver has a fair dice with 4 faces numbered 1 to 4
 He also has a fair dice with 6 faces numbered 1 to 6
 He draws this possibility diagram to represent the outcomes of rolling both dice.

4-sided dice	4		x	x	x	x	x	x
	3		x	x	x	x	x	x
	2		x	x	x	x	x	x
	1		x	x	x	x	x	x
			1 2 3 4 5 6					
			6-sided dice					

Find the probability of rolling a different number on each dice.

..... [1]

- 16** Solve.

$$8x = 52 - 4(x - 2)$$

$x =$ [2]

17 Pierre wants to investigate the favourite hobbies of students in his year group at school.

- (a) There are 100 girls and 120 boys in his year group at school.
Pierre decides to sample 20 girls and 24 boys rather than the same number of girls and boys.

Write down a reason to support Pierre's decision.

.....
..... [1]

- (b) Pierre decides to give the students in his sample a questionnaire instead of interviewing them.

Write down a reason why this may **not** be a good decision.

.....
..... [1]

- (c) Pierre wants to use a fair sampling method to randomly select the 20 girls.

Put a tick (✓) next to the sampling method that is the most likely to be random and fair.

Choose the youngest 20 girls.

☐

Choose the first 20 girls he sees coming into school.

☐

Put the names of all the girls in a hat and pick 20 names out of the hat without looking.

☐

One class in the school has 20 girls in it. Choose all of the girls in this class.

☐

[1]

- 18** Yuri has two boxes containing red, black and blue pens only.

Box A contains 3 red pens, 5 black pens and 2 blue pens.

Box B contains 2 red pens, 2 black pens and 1 blue pen.

Yuri picks a pen from one of the boxes at random.

- (a)** Yuri thinks the probability that the pen will be green is 0

Explain why Yuri is correct.

.....
 [1]

- (b)** Yuri says,

‘I am more likely to pick a red pen from box A than from box B
 because there are more red pens in box A than box B.’

Explain why Yuri is **not** correct.

.....
 [1]

- 19** The edge of a square has a length of 23.8 cm.

The perimeter of this square is equal in length to the circumference of a circle.

Work out the radius of the circle.

..... cm [3]

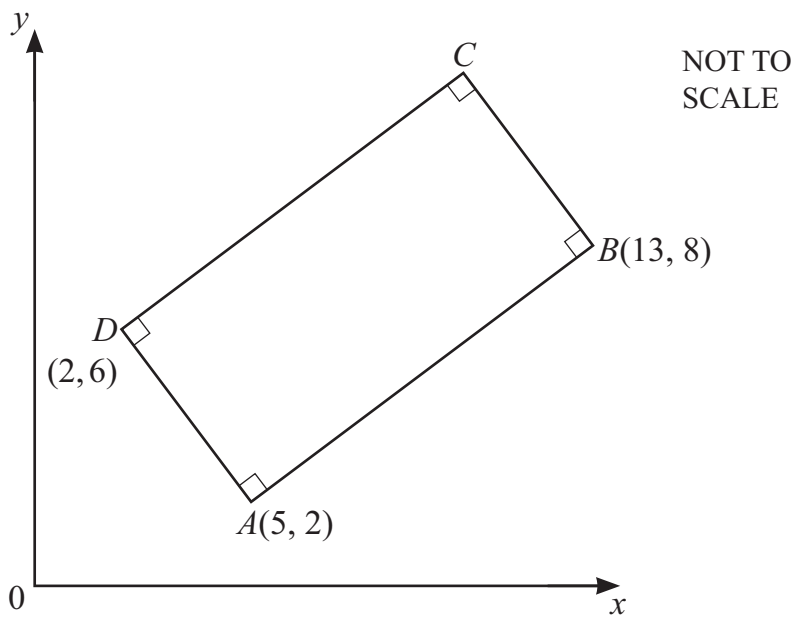
- 20 An expression for the area of this rectangle is $12h^2 + 15h$



Find an expression for the length of the rectangle.
Give your answer in its simplest form.

..... [1]

- 21 The diagram shows a rectangle $ABCD$ drawn on coordinate axes.



Work out the midpoint of line AC .

(..... ,) [2]

- 22** Samira has a bag containing coloured balls.
Some of the balls are red.
The number of balls in the bag that are **not** red is four times the number of red balls.
Samira picks a ball at random from the bag.

Work out the probability the ball is red.

..... [1]

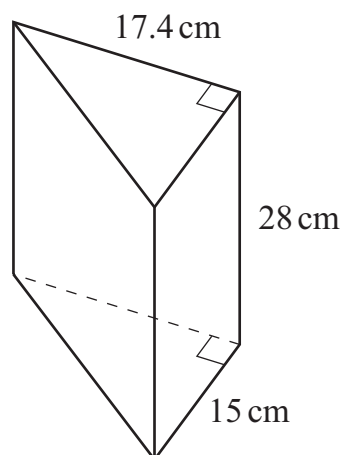
- 23** Here are the first four terms of a linear sequence.

3.1, 2.3, 1.5, 0.7

Find the n th term of this sequence.

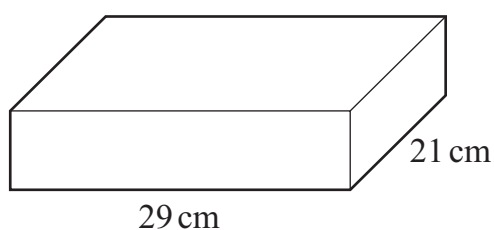
..... [2]

- 24 The diagram shows a container in the shape of a triangular prism.
The cross section is a right-angled triangle.
The container is completely filled with water.



NOT TO
SCALE

The water is then poured into the cuboid-shaped tray in the second diagram.
The water completely fills the tray.



NOT TO
SCALE

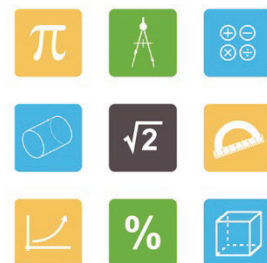
Find the height of the tray.

..... cm [4]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced annually and is available to download at <https://lowersecondary.cambridgeinternational.org/>

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.



Mathematics

Stage 8

Paper 2

2023

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Brackets in mark scheme

When brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6(58...)	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30.....cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e. 18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

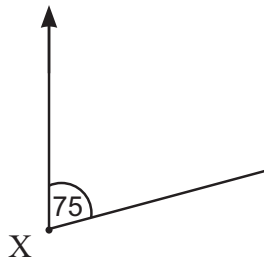
The table shows acceptable and unacceptable versions of various answers.

For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

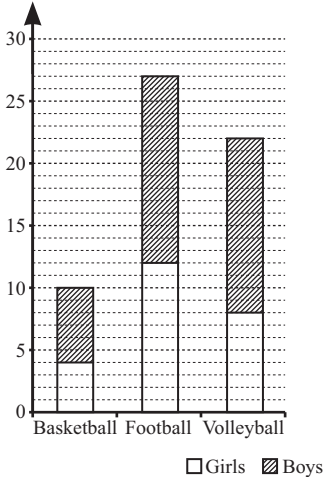
The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance
1	(\$)7650	2	Award 1 mark for 9000×0.85 or equivalent or 1350	
2	Any two equal angles in the range $90^\circ < k < 180^\circ$	1		k does not need to be an integer.
3	(A =) rational numbers (B =) integers (C =) natural numbers	1		All three answers correct for the mark.
4	Correct diagram showing bearing of 075° 	1		Accept $\pm 2^\circ$
5	(decreasing) 2013 increasing 20 2018	2	Award 1 mark for two or three correct sentences.	Accept equivalent alternatives for increasing, e.g. rising.
6(a)	$\begin{pmatrix} -2 \\ 4 \end{pmatrix}$	1		
6(b)	Triangle drawn at (6, 5), (9, 6), (8, 8).	1		

Question	Answer	Marks	Part Marks	Guidance
7	True False ☒ ☐ ☒ ☐ ☒ ☐ ☒ ☐	2	Award 1 mark for two or three correct answers.	Accept any clear indication.
8(a)	22 : 9	2	Award 1 mark for partially simplified ratio, e.g. 550 : 225 110 : 45 22 cm : 9 cm	Accept units in ratio for 1 mark, but inconsistent units scores 0 marks, e.g. 110 cm : 45 m.
8(b)	5 : 2 : 7	1		
9(a)	$x = r - 5$ final answer	1		
9(b)	$x = y(2p + 3) - 7$ or $x = 2py + 3y - 7$ final answer	2	Award 1 mark for $y(2p + 3) = 7 + x$ or better	Or better, e.g. $2py + 3y = 7 + x$
10	806 000 0.0701 4.00	2	Award 1 mark for two correct answers.	Do not accept 0.07010(0) Do not accept 4(.0)

Question	Answer	Marks	Part Marks	Guidance												
11	<table><tr><td>x</td><td>(-2)</td><td>-1</td><td>(2)</td></tr><tr><td>y</td><td>5</td><td>(3)</td><td>-3</td></tr></table> and Correct ruled graph extending at least between (-2, 5) and (2, -3).	x	(-2)	-1	(2)	y	5	(3)	-3	3	Award 2 marks for correctly completed table. or Award 1 mark for two correct values in the table or correctly plotting <i>their</i> three coordinates.					
x	(-2)	-1	(2)													
y	5	(3)	-3													
12	<table><tr><td></td><td>On Angelique's</td><td>Not on Angelique's</td></tr><tr><td>Below $y = 10$</td><td>(-1, -5)</td><td>(5, 1)</td></tr><tr><td>On $y = 10$</td><td>(2, 10)</td><td>(5, 10)</td></tr><tr><td>Above $y = 10$</td><td>(2.5, 12.5)</td><td>(-10, 50)</td></tr></table>		On Angelique's	Not on Angelique's	Below $y = 10$	(-1, -5)	(5, 1)	On $y = 10$	(2, 10)	(5, 10)	Above $y = 10$	(2.5, 12.5)	(-10, 50)	2	Award 1 mark for three or four coordinates correct out of five.	
	On Angelique's	Not on Angelique's														
Below $y = 10$	(-1, -5)	(5, 1)														
On $y = 10$	(2, 10)	(5, 10)														
Above $y = 10$	(2.5, 12.5)	(-10, 50)														
13	140(°)	2	Award 1 mark for 360 ÷ (3 + 6 + 7 + 2)	1 mark implied by 20 or for answer 40, 60 or 120												

Question	Answer	Marks	Part Marks	Guidance
14	Correct chart with key  ☐ Girls ☒ Boys	3	Award 2 marks for a correct bar chart with no key. or Award 1 mark for correct key and Award 1 mark for a bar chart with two correct columns.	For 1, 2 or 3 marks: The order of the girls and boys bars can be reversed, but this must be consistent. Bars must be consistent widths. Any distinct shading/colours may be used for shading and key.
15	$\frac{20}{24}$	1		Accept equivalent fractions, decimals or percentage.
16	$(x =) 5$	2	Award 1 mark for $8x = 52 - 4x + 8$ or better or answer of $\frac{11}{3}$ or equivalent following $8x = 52 - 4x - 8$	

Question	Answer	Marks	Part Marks	Guidance
17(a)	A correct reason, e.g. The sample will be in proportion to the population.	1		Accept reasons implying understanding of the proportion, e.g. - There are more boys in his year group so there should be more boys in the sample. - His sample will be representative of the population in terms of gender.
17(b)	A correct reason, e.g. - They may not return a questionnaire. - You cannot explain questions in a questionnaire. - It may take longer to process the results. - They may not answer all the questions.	1		
17(c)	Choose the youngest 20 girls. ☐ Choose the first 20 girls he sees coming into school. ☐ Put the names of all the girls in a hat and pick 20 names out of the hat without looking. ☒ One class in the school has 20 girls in it. Choose all of the girls in this class. ☐	1		Accept any clear indication.

Question	Answer	Marks	Part Marks	Guidance
18(a)	A correct explanation, e.g. Probability of 0 means impossible and there are no green pens in either box.	1		Accept 'there are no green pens in either box' for the mark.
18(b)	A correct explanation, e.g. Probability of red from box A is 30% and from box B is 40%.	1		Accept any equivalent explanation, e.g. If you double the number of each pen in box B there would be 4 out of 10 which is more than 3 out of 10 Accept brief explanation, e.g. $\frac{2}{5} > \frac{3}{10}$ Do not accept an incomplete explanation, e.g. there are more pens in total in box A.
19	15.1 to 15.2 (cm)	3	Award 2 marks for complete method, e.g. $23.8 \times 4 \div (2 \times \pi)$ or Award 1 mark for 23.8×4 or <i>their</i> perimeter = $2\pi r$ or better.	Accept value of π between 3.14 and $\frac{22}{7}$ For 3 marks, accept 15 with correct working. 1 mark implied by 95.2 Or better, e.g. <i>their</i> perimeter $\div (2 \times \pi)$
20	$4h + 5$ final answer	1		

Question	Answer	Marks	Part Marks	Guidance
21	(7.5, 7)	2	Award 1 mark for one correct coordinate or for $C = (10, 12)$.	Accept equivalents for 7.5
22	$\frac{1}{5}$	1		Accept 20% or 0.2 Do not accept 1 in 5 or 1:5
23	$3.9 - 0.8n$	2	Award 1 mark for $c - 0.8n$ or $3.9 + kn$ where $k < 0$ or $3.1 + p(n - 1)$ where $p < 0$	Accept equivalent correct answers.
24	6 (cm)	4	Award 3 marks for $0.5 \times 17.4 \times 15 \times 28 = 29 \times 21 \times h$ or better. or Award 2 marks for correct method to find volume of triangular prism, e.g. $0.5 \times 17.4 \times 15 \times 28$ or Award 1 mark for - Area of the triangle $\frac{1}{2} \times 17.4 \times 15$ - Volume of the rectangular prism $17.4 \times 15 \times 28$ - Volume of the cuboid-shaped tray $29 \times 21 \times h$ <i>Their</i> triangular prism volume $\div (29 \times 21)$	Or better, e.g. $0.5 \times 17.4 \times 15 \times 28 \div (29 \times 21)$ 2 marks implied by 3654 1 mark implied by 130.5 or 7308

BLANK PAGE